

2nd Day Internship [8th June 2026] CoE - MathWorks DeepTech Labs

https://in.mathworks.com/help/matlab/index.html?s_tid=hc_panel

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Semester & Branch: VII Sem ECE 2023-27

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```
RR_12=5
```

```
RR_12 = 5
```

```
clc=5
```

```
clc = 5
```

```
func=666
```

```
func = 666
```

```
namelengthmax
```

```
ans = 2048
```

```
exist aabb
```

```
ans = 0
```

```
exist display
```

```
ans = 5
```

```
% check to see if A is greater than  
B A=10; B=20; if A > B "greater"  
elseif A < B  
"less"  
end
```

```
ans = "less"
```

```
round(100*rand(1,10))
```

```
ans = 1×10  
71 3 28 5 10 82 69 32 95 3
```

true or false

```
A = true(3,4)
```

```
A = 3x4 logical array
```

```
1 1 1 1
1 1 1 1
1 1 1 1
```

```
false(4,6)
```

```
ans = 4x6 logical array
```

```
0 0 0 0 0 0
0 0 0 0 0 0
0 0 0 0 0 0
0 0 0 0 0 0
```

```
magic(2)
```

```
ans = 2x2
```

```
1
3
4
2
```

```
tm = magic(5)
```

```
tm = 5x5
```

```
17 24 1 8 15
23 5 7 14 16
4 6 13 20 22
10 12 19 21 3 11 18
25 2 9
```

```
tm1=tm(1:2,1:2)
```

```
tm1 = 2x2
```

```
17 24
23 5
```

```
tm2=tm(:,3:4)
```

```
tm2 = 5x2
```

```
1 8
7 14
13 20
19 21
25 2
```

```
true(3,2,2)
```

```
ans = 3x2x2 logical array ans(:,:,1)
```

```
=
```

```
1 1
```

```
1 1
1 1
```

```
ans(:, :, 2) =
```

```
1 1
1 1
1 1 ans
= 1
```

greater logical array of selected sparsity

```
A = logical(sparse(5,3));
whos A
```

Name	Size	Bytes	Class	Attributes
A	5x3	41	logical	sparse

```
F = false(4,like=A);
whos F
```

Name	Size	Bytes	Class	Attributes
F	4x4	49	logical	sparse

create grids

```
y = linspace(0,100)
```

```
y = 1x100
0 1.0101 2.0202 3.0303 4.0404 5.0505 6.0606 7.0707 ...
```

```
y = linspace(0,10,100)
```

```
y = 1x100
0 0.1010 0.2020 0.3030 0.4040 0.5051 0.6061 0.7071 ...
```

log space

```
y = logspace(5,4)
```

```
y = 1x50
105 x
1.0000 0.9541 0.9103 0.8685 0.8286 0.7906 0.7543 0.7197 ...
```

```
y = logspace(5,4,100)
```

```
y = 1x100
105 x
1.0000 0.9770 0.9545 0.9326 0.9112 0.8902 0.8697 0.8498 ...
```

```
y = logspace(5,3.14)
```

y = 1×50
10⁵ ×

1.0000	0.9163	0.8396	0.7693	0.7050	0.6460	0.5919	0.5424 ...
--------	--------	--------	--------	--------	--------	--------	------------

```
Y = logspace(5,3.14,100)
```

Y = 1×100
10⁵ ×

1.0000	0.9577	0.9171	0.8783	0.8411	0.8055	0.7714	0.7387 ...
--------	--------	--------	--------	--------	--------	--------	------------

Frequency space

```
[f1,f2] = freqspace(5)
```

f1 = 1×5
-0.8000 -0.4000 0 0.4000 0.8000

f2 = 1×5
-0.8000 -0.4000 0 0.4000 0.8000

```
[f1,f2] = freqspace([0 5])
```

f1 = 1×5
-0.8000 -0.4000 0 0.4000 0.8000

f2 = 1×0 empty double row vector

mesh grid 2D

```
x = 1:3;
y = 1:5;
[X,Y] = meshgrid(x,y)
```

X = 5×3

1	2	3
1	2	3
1	2	3
1	2	3
1	2	3

Y = 5×3

1	1	
2	2	
3	3	
4	4	
5	5	

MESH GRID 3D

```
x = 0:2:6;
y = 0:1:6;
z = 0:3:6;
[X,Y,Z] = meshgrid(x,y,z);
```

size of array

```
aa=magic(7)
```

```
aa = 7x7
    30    39    48     1    10    19    28
    38    47     7     9    18    27    29
    46     6     8    17    26    35    37
     5    14    16    25    34    36    45
    13    15    24    33    42    44     4
    21    23    32    41    43     3    12
    22    31    40    49     2    11    20
```

number of elements in 3D matrix

```
A = magic(4);
A(:,:,2) = A'
```

```
A =
A(:,:,1) =
    16     2     3    13
     5    11    10     8
     9     7     6    12
     4    14    15     1
```

```
A(:,:,2) =
    16     5     9     4
     2    11     7    14
     3    10     6    15
    13     8    12     1
```

determine scalar and vector from matrix

```
A = [1 2; 3 4];
TF = isscalar(A)
```

```
TF = logical
0
```

```
A = [1 2; 3 4];
TF = isvector(A)
```

```
TF = logical
0
```

is matrix

```
A3 = zeros(1,3,2); TF
= ismatrix(A3)
```

```
TF = logical
     0 is row and
column V =
rand(5,1); tf
= isrow(V)
```

```
tf = logical
     0
```

```
V = rand(5,1);
tf = iscolumn(V)
```

```
tf = logical
     1
```

resize

```
aa=magic(16)
```

```
aa = 16×16
    256     2     3    253    252     6     7    249    248    10    11    245    244 ...
    17    239    238     20     21    235    234     24     25    231    230     28     29
    33    223    222     36     37    219    218    40    41    215    214    44    45
   208     50     51    205    204     54     55    201    200     58     59    197    196
    192     66     67    189    188     70     71    185    184     74     75    181    180
     81    175    174     84     85    171    170     88     89    167    166     92     93
     97    159    158    100    101    155    154    104    105    151    150    108    109
   144    114    115    141    140    118    119    137    136    122    123    133    132
    128    130    131    125    124    134    135    121    120    138    139    117    116
   145    111    110    148    149    107    106    152    153    103    102    156    157
    161     95     94    164    165     91     90    168    169     87     86    172    173
     80    178    179     77     76    182    183     73     72    186    187     69     68
     64    194    195     61     60    198    199     57     56    202    203     53     52
   209     47     46    212    213     43     42    216    217     39     38    220    221
    225     31     30    228    229     27     26    232    233     23     22    236    237
     ⋮
     ⋮
```

```
head(aa',2)'
```

```
ans = 16×2
    256     2
     17    239
     33    223
    208     50
    192     66
     81    175
     97    159
   144    114
```

```

128 130
145 111
161 95
80 178
64 194
209 47
225 31
⋮

```

resize vector

```

A = [1;3;5;7];
B1 = resize(A,2)

```

```

B1 = 2×1
1

```

```

3

```

Reshaping

```

A = 1:10;
B = reshape(A,[5,2])

```

```

B = 5×2
1     6
2     7
3     8
4     9
5    10

```

permute

```

rng default A =
rand(3,4,2)

```

```

A =
A(:,:,1) =
    0.8147    0.9134    0.2785    0.9649
    0.9058    0.6324    0.5469    0.1576
    0.1270    0.0975    0.9575    0.9706

```

```

A(:,:,2) =
    0.9572    0.1419    0.7922    0.0357
    0.4854    0.4218    0.9595    0.8491
    0.8003    0.9157    0.6557    0.9340

```

```

B = permute(A,[3 2 1])

```

```

B =
B(:,:,1) =

```

0.8147	0.9134	0.2785	0.9649
0.9572	0.1419	0.7922	0.0357

B(:, :, 2) =

0.9058	0.6324	0.5469	0.1576
0.4854	0.4218	0.9595	0.8491

B(:, :, 3) =

0.1270	0.0975	0.9575	0.9706
0.8003	0.9157	0.6557	0.9340

rearranging

```
aa = magic(4)
```

```
aa = 4x4      16      2
      3      13
           5      11      10      8
           9      7      6      12
      4      14      15      1
```

```
sort(aa)
```

```
ans = 4x4      4      2
      3      1
      5      7      6      8
           9      11      10      12
          16      14      15      13
```

```
A = magic(4)
```

```
A = 4x4
      16      2      3      13
           5      11      10      8
           9      7      6      12
           4      14      15      1
```

```
B = A.'
```

```
B = 4x4
      16      5      9      4
      2      11      7      14
      3      10      6      15
      13      8      12      1
```

c Transpose

```
A = [2 1; 9 7; 2 8; 3 5]
```

```
A = 4x2
      2      1
      9      7
      2      8
```


3 5

```
B = A'
```

B = 2×4

2	9	2	3
1	7	8	5

circ shift

```
A = (1:10)'
```

A = 10×1

1
2
3
4
5
6
7
8
9
10

```
Y = circshift(A,3)
```

Y = 10×1

8
9
10
1
2
3
4
5
6
7

uint 32

```
y = uint32(10);  
whos y
```

Name	Size	Bytes	Class	Attributes	y
1x1	4	uint32			

double precision arrays

```
x = 10;  
whos x
```

Name	Size	Bytes	Class	Attributes	x
1x1	8	double			

is NAN

```
A = 0./[-2 -1 0 1 2]
```

```
A = 1x5
    0    0 NaN    0    0
```

```
TF = isnan(A)
```

```
TF = 1x5 logical array
    0    0    1    0    0
```

```
str = "i want you take i want"
```

```
str = "i want you take i
want"
```

```
str=split(str)
```

```
str = 6x1 string
    "i"
    "want"
    "you"
    "take"
    "i"
    "want"
```

```
str = ["Thanu","Adi";      "tharun","swathi";
"cherry","sharmila"]
```

```
str = 3x2 string
    "Thanu"    "Adi"
    "tharun"    "swathi"
    "cherry"    "sharmila"
```

```
str = join(str)
```

```
str = 3x1 string
    "Thanu Adi"
    "tharun swathi"
    "cherry shar..."
```

```
s1 = ["Adi is very intelligent boy","he is","LT509"]
```

```
s1 = 1x3 string
    "Adi is very intelligent..." "he is"    "LT509"
```

```
p1 = "Adi"
```

```
p1 = "Adi"
```

```
contains(s1,p1)
```

```
ans = 1x3 logical array    1
    0    0
```

```
DateStrings = {'2014-05-26'; '2014-08-03'}; t = datetime(DateStrings, 'InputFormat', 'yyyy-MM-dd')
```

```
t = 2×1 datetime 26-May2014  
03-Aug-2014
```

Data strings

```
DateStrings =  
{'2014-05-26T13:30-05:00'; '2014-08-26T13:30-04:00'; '2014-09-  
26T13:30Z'}
```

```
DateStrings = 3×1 cell  
'2014-05-26T13:30-05:00'  
'2014-08-26T13:30-04:00'  
'2014-09-26T13:30Z'
```

convert excel data into date

```
X = [39558, 39600; 39700, 39800]
```

```
X = 2×2  
    39558    39600  
    39700    39800
```

```
t = datetime(X, 'ConvertFrom', 'excel')
```

```
t = 2×2 datetime  
20-Apr-2008 01-Jun-2008  
09-Sep-2008 18-Dec-2008
```

Structures

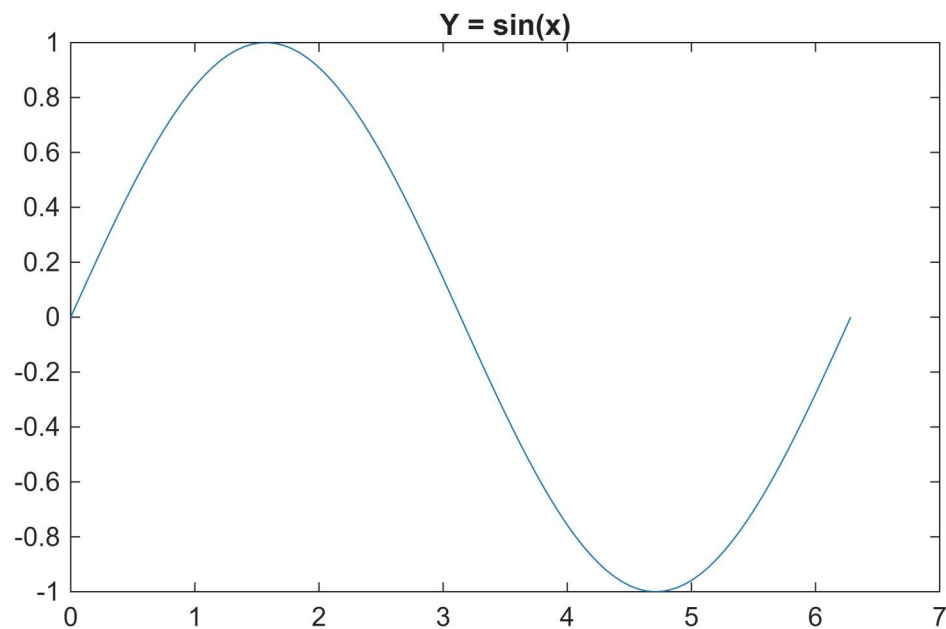
```
data.x = linspace(0,2*pi);  
data.y = sin(data.x);  
data.title = 'Y = sin(x)'
```

```
data = struct with fields:    x: [0 0.0635 0.1269 0.1904 0.2539 0.3173 0.3808 0.4443 0.5077 0.5712 0.6347  
0.6981 0.7616 0.8251 0.8885 0.95    y: [0 0.0634 0.1266 0.1893 0.2511 0.3120 0.3717 0.4298 0.4862 0.5406 0.5929  
0.6428 0.6901 0.7346 0.7761 0.81    title: 'Y = sin(x)'
```

```
zz = data.x
```

```
zz = 1×100  
    0    0.0635    0.1269    0.1904    0.2539    0.3173    0.3808    0.4443 ...
```

```
plot(data.x,data.y)  
title(data.title)
```



```
field = 'f'
```

```
field = 'f'
```

```
value = {'some text';  
[10, 20, 30];  
magic(5)}; s =  
struct(field,value)
```

```
s = 3x1 struct
```

Fields	f
1	'some text'
2	[10,20,30]
3	5x5 double

structures with multiple fields

```
field1 = 'f1'; value1 = zeros(1,10); field2 = 'f2'; value2 =  
{'a', 'b'}; field3 = 'f3'; value3 = {pi, pi.^2}; field4 = 'f4';  
value4 = {'fourth'}; s =  
struct(field1,value1,field2,value2,field3,value3,field4,value4)
```

```
s = 1x2 struct
```

Fields	f1	f2	f3	f4
1	[0,0,0,0,0,0,0,0,0,0]	'a'	3.1416	'fourth'
Fields	f1	f2	f3	f4

2	[0,0,0,0,0,0,0,0,0]	'b'	9.8696	'fourth'
---	---------------------	-----	--------	----------

s(1)

```
ans = struct with fields:
f1: [0 0 0 0 0 0 0 0 0]
f2: 'a'
f3: 3.1416
f4: 'fourth'
```

s(2)

```
ans = struct with fields:      f1:
[0 0 0 0 0 0 0 0 0]
f2: 'b'
f3: 9.8696      f4:
'fourth'
```

Functions

```
fun = 'round'; x1
= pi; y =
feval(fun,x1)
```

y = 3

```
products = ["Tomato" "carrot" "Mango" "Mushroom"]
```

```
products = 1x4 string
"Tomato"      "carrot"      "Mango"      "Mushroom"
```

```
Prices = [1 0.5 2.50 1.99]; d =
dictionary(products,Prices)
```

d = dictionary (string $\mapsto$ double) with 4

entries:

```
"Tomato"       $\mapsto$  1
"carrot"       $\mapsto$  0.5000
"Mango"        $\mapsto$  2.5000
"Mushroom"     $\mapsto$  1.9900
```

Time series

```
ts1 = timeseries([5 10 15 20 25]','"Name","MyTimeSeries")
```

timeseries

Common Properties:

```
Name: 'MyTimeSeries'
Time: [5x1 double]
TimeInfo: [1x1 tsdata.timemetadata]
```

```
Data: [5x1 double]
DataInfo: [1x1 tsdata.datametadata]
More properties, Methods
```

```
ts2 = timeseries(rand(2,5))
```

```
timeseries
```

```
Common Properties:
```

```
    Name: 'unnamed'
    Time: [2x1 double]
TimeInfo: [1x1 tsdata.timemetadata]
    Data: [2x5 double]
DataInfo: [1x1 tsdata.datametadata]
```

```
More properties, Methods
```